Wax Print Image Processor

Setup & User Guide for Windows

A step-by-step guide for first-time users

Overview

This guide walks you through everything you need to do to get the Wax Print Image Processor running on your Windows computer — from installing the required software to running your first analysis. No prior programming experience is required.

The application processes gecko bite print images through several stages:

- Match — automatically locates bite mark patterns in your images
- Manual — lets you review and correct the automated results
- FitProj — processes the collected data and identifies the central tooth
- Format — combines data across multiple images and creates visualizations
- Analyze — an interactive tool to explore your results

Part 1: One-Time Setup

You only need to complete this section once. Once everything is installed, you can skip directly to Part 2 for future sessions.

Step 1: Install Python

The application is written in Python 3.10. You need to install Python before anything else will work.

1. Open your web browser and go to: https://www.python.org/downloads/release/python-31010/
2. Scroll down and click "Windows installer (64-bit)" to download the installer.
3. Once downloaded, double-click the installer file to open it.
4. **IMPORTANT:** Before clicking "Install Now", make sure to check the box that says **"Add Python to PATH"** at the bottom of the installer window. This is crucial.
5. Click "Install Now" and wait for the installation to finish.

6. Click "Close" when done.

✅ **Tip:** To verify Python installed correctly: press the Windows key, type 'cmd', press Enter, then type 'python --version' and press Enter. You should see something like 'Python 3.10.10'.

Step 2: Install Git

Git is a tool that lets you download code from GitHub. You need it to get the application files onto your computer.

1. Go to: https://git-scm.com/download/win in your web browser.
2. The download should start automatically. If not, click "Click here to download manually."
3. Run the downloaded installer. You can click "Next" through all the options — the defaults are fine.
4. Click "Finish" when the installation is complete.

Step 3: Download the Application

Now you will download the application code from GitHub onto your computer.

7. Decide where you want to store the application. A good choice is your Documents folder.
8. Open File Explorer (the folder icon on your taskbar) and navigate to that folder.
9. Click in the address bar at the top of File Explorer (where it shows the folder path), type 'cmd', and press Enter. A black Command Prompt window will open.
10. In the Command Prompt, type the following exactly and press Enter:

```
git clone https://github.com/ecytryn/Wax-Print-Image-Processor.git
```

11. Wait for it to finish. You will see text scrolling by. When it stops and shows a prompt again, the download is complete.
12. A new folder called "Wax-Print-Image-Processor" will appear in your chosen location.

Step 4: Set Up the Python Environment

Python applications use "virtual environments" to keep their required packages isolated. This prevents conflicts with other software on your computer. You only need to do this once.

5. In the same Command Prompt window (or open a new one inside the Wax-Print-Image-Processor folder using the same method as Step 3), type the following and press Enter:

```
cd Wax-Print-Image-Processor
```

This navigates into the application folder.

6. Create the virtual environment by typing the following and pressing Enter:

```
python -m venv venv
```

📌 **Note:** If you see an error, try 'python3 -m venv venv' instead. Wait until the prompt returns — this may take a minute.

7. Activate the virtual environment by typing:

```
venv\Scripts\activate.bat
```

You will know it worked when you see "(venv)" appear at the beginning of the line in your Command Prompt, like this:

```
(venv) C:\Users\YourName\Documents\Wax-Print-Image-Processor>
```

8. Now install all the required packages. Type the following and press Enter:

```
pip install -r requirements.txt
```

This will download and install all the necessary components. It may take several minutes. When it finishes and the prompt returns, you are ready.

📌 **Note:** If you see a notice about upgrading pip, run: pip install --upgrade pip — then run the pip install -r requirements.txt command again.

Step 5: Initialize the Application Folder Structure

Run this command once to create the folders the application needs:

```
python main.py
```

This creates an "/img" folder (where you will put your images) and several output folders. You are now fully set up!

Part 2: Starting the Application (Every Session)

Each time you want to use the application, follow these steps to get it running again.

Step 1: Open Command Prompt in the Application Folder

13. Open File Explorer and navigate to your "Wax-Print-Image-Processor" folder.
14. Click in the address bar, type 'cmd', and press Enter.

Step 2: Activate the Virtual Environment

Type the following and press Enter:

```
venv\Scripts\activate.bat
```

You should see "(venv)" appear at the start of the line. The application is now ready to use.

> 📌 **Note:** If you forget to activate the virtual environment, the application will fail with an error about missing modules. Always activate it first!

Part 3: Adding Your Images

Before running any analysis, you need to place your gecko bite print images in the right location.

15. Inside the "Wax-Print-Image-Processor" folder, find the folder named "img" (it was created in Part 1, Step 5).
16. Copy or move your image files into this "img" folder.

✅ **Tip:** Images are processed in alphabetical order, not by date. Consider naming your files with dates (e.g., 2024-01-15_gecko01.jpg) to keep them in chronological order.

Part 4: Running the Application

All commands follow this basic format:

```
python main.py -n [number of images] -s [starting position] [process name]
```

Where:

- -n specifies how many images to process
- -s specifies which image to start from (counting from 0, so the first image is 0, the second is 1, etc.)
- [process name] is the step you want to run (match, manual, fitproj, format, or analyze)

Common Commands

Process all images from the beginning (runs match, manual, fitproj, and format):

```
python main.py
```

Run just the template matching step on the first image:

```
python main.py -n 1 -s 0 match
```

Run manual review on a single image (the 3rd image in your folder):

```
python main.py -n 1 -s 2 manual
```

Process all 10 images starting from the first:

```
python main.py -n 10 -s 0 match fitproj format
```

Open the interactive analysis tool:

```
python main.py analyze
```

📌 **Note:** The 'analyze' step automatically runs 'format' first, so you do not need to run format separately before analyzing.

Part 5: Using the Manual Review Tool

When you run the "manual" step, a window will open showing your image. Use the following controls:

Key / Action	What it does
`Left click`	Plot or remove a data point
`Tab or 1/2/3/4`	Switch between: Tooth mode, Gap mode, Center Tooth mode, Center Gap mode
`Space`	Toggle viewing mode (hides boxes so you can see the image more clearly)
`S`	Save your changes and close the window
`Esc`	Close the window without saving
`← Left arrow`	Save, close, and go back to the previous image
`→ Right arrow`	Save, close, and move to the next image

Part 6: Using the Analyze Tool

When you run the "analyze" step, an interactive window opens for exploring your results.

- Left click on a point — opens the projected image for that day
- Left click and drag — opens projected images for all selected days stacked together
- Right click — opens the manual review tool for that day's data (note: changes won't appear in the plots until you run 'format' or 'analyze' again)

✅ **Tip:** To force-quit the application at any time, press Ctrl+C in the Command Prompt window, then close any open image windows.

Troubleshooting

"python is not recognized as an internal or external command"

Python was not installed correctly, or the "Add Python to PATH" option was not checked during installation. Try uninstalling Python and reinstalling it, making sure to check that box.

"ModuleNotFoundError" when running the app

The virtual environment is probably not activated. Make sure you see "(venv)" at the start of your command line. If not, run:

```
venv\Scripts\activate.bat
```

"No such file or directory: img"

The img folder does not exist yet. Run the following command to create it:

```
python main.py
```

The application opens but nothing happens

Check that you placed image files in the "img" folder and that your -n and -s values are correct for the number of images you have.

A window appears and immediately closes

Look at the Command Prompt for an error message. If you see a red traceback, take a screenshot and send it to the developer for help.

Part 7: Working with Multiple Animals

The application is designed to work with one set of images at a time, stored in its TemplateMatching/img/ folder. If you are working with images from multiple animals, the

recommended approach is to keep each animal's images and results in their own storage folders outside the repository, and copy them in and out as needed.

Recommended Folder Structure

Create a storage folder somewhere outside the repository — for example, inside your Documents folder. Give each animal its own subfolder containing an img subfolder for the raw images and a processed subfolder where results will be saved:

```
Documents/GeckoTeethResearch/Animals/
    animal1/
        img/          (your raw images live here permanently)
        processed/    (analysis results will be stored here)
    animal2/
        img/
        processed/
```

Workflow for Each Animal

Follow these four steps each time you want to process images for a new animal. Your original images are never deleted — you are only ever working with copies inside the repository.

- **Copy the animal's images into the repository.** Open File Explorer, navigate to Animals/animal1/img/, select all image files, and copy them (Ctrl+C). Then navigate to Wax-Print-Image-Processor/TemplateMatching/img/ and paste them (Ctrl+V). Your originals remain safely in the Animals folder.
- **Run the analysis.** Follow Parts 2–6 of this guide as normal. Results will be written into Wax-Print-Image-Processor/TemplateMatching/processed/.
- **Move the results to the animal's storage folder.** Navigate to Wax-Print-Image-Processor/TemplateMatching/processed/, select all contents, cut them (Ctrl+X), then navigate to Animals/animal1/processed/ and paste (Ctrl+V). This moves the results out of the repository into permanent storage.
- **Delete the image copies from the repository.** Navigate to Wax-Print-Image-Processor/TemplateMatching/img/, select all image files, and delete them. Since you copied (not moved) in Step 1, your originals are untouched in the Animals folder. The repository is now clean and ready for the next animal.

Note: If you need to re-analyze a previous animal later, simply repeat from Step 1 — copy their images back into the repository, run the analysis again, and move the new results back out to their storage folder.

Quick Reference Card

Cut out or print this section for easy reference during your sessions.

Every Session Checklist

9. Open Command Prompt inside the Wax-Print-Image-Processor folder

10. Activate environment: venv\Scripts\activate.bat
11. Place images in the /img folder
12. Run your desired command (see below)

Command Reference

```
python main.py                                (run all steps on all images)
python main.py -n 1 -s 0 match               (template match, image 1)
python main.py -n 1 -s 0 manual              (manual review, image 1)
python main.py -n 1 -s 0 fitproj             (fit/project, image 1)
python main.py -n 5 -s 0 match fitproj       (match + fitproj, first 5
images)
python main.py format                         (format/visualize all data)
python main.py analyze                        (open interactive analysis
tool)
```

For further help, contact the developer at: https://github.com/marclai2003/Wax-Print-Image-Processor